

Jayashini Jayaweera

071 9234816 | jayashinijayaweera@gmail.com | <https://www.linkedin.com/in/jayashini-jayaweera/> | <https://github.com/Jayashini>

ABOUT ME

Information Systems undergraduate focused on cybersecurity, software engineering, and reliable system design. I build full-stack applications and enjoy digging into how systems actually work - from database design to identifying and fixing security weaknesses. Looking for an internship where I can apply this hands-on approach and grow alongside an engineering team.

EDUCATION

University of Colombo School of Computing

BSc. in Information Systems

GPA 3.36

Graduating 2027

Kalutara Balika Vidyalaya

GCE Advanced Level Examinations, Physical Science Stream

2022

TECHNICAL SKILLS

Languages: C/C++, Python, PHP, JavaScript, Typescript, Go

Frontend: React, Preact, Tailwind CSS

Tools: Git, Docker, VS Code, Burp Suite

Networking: TCP/IP, Subnetting, OSI Model, HTTP/HTTPS, DNS

Backend: Node.js, Next.js

Database: MySQL, MongoDB, RDBMS, SQL Queries, Transactions, Normalization, ACID Properties

PROJECTS

FiremeX | *Go, Preact, PostgreSQL, Docker, open CV, YOLO, Github, ML Flow*

June 2026 – Present

- Developing an AI-driven, real-time fire detection system for CCTV platforms, leveraging computer vision (OpenCV/YOLO) and Python middleware to minimize manual monitoring errors.
- A scalable backend using Go and PostgreSQL to handle real-time video stream analysis, ensuring sub-5-second alert latency for critical fire and smoke event detection.
- Preact.js for a centralized monitoring dashboard, providing security officers with real-time incident tracking, risk assessment, and role-based access control (RBAC).
- Engineered a MySQL database schema with 25+ relational tables covering user management, tournaments, transactions, and dynamic sport-specific result fields, while implementing security best practices including password hashing.

Colombo Sports E - Portal | *PHP, MySQL, JS, CSS, MVC Architecture, REST API*

August 2025 – April 2026

- Developed UOC Sports Management System, a full-stack web application to manage university sports administration, including facility reservations, equipment tracking, and attendance monitoring.
- Designed and implemented a role-based access control system supporting seven user roles (Admin, Student, Captain, Coach, Sport Manager, Equipment Manager, Registrar), enabling secure, permission-specific dashboards and workflows.
- Optimized operational efficiency and PDPA compliance through centralized data authority, resulting in streamlined vessel tracking and enhanced business intelligence.

Data Governance Framework | *Magnate Shipping Agency*

November 2025 – February 2026

- Designed a comprehensive Data Governance and Management Framework for Magnate Shipping Agency to modernize maritime logistics.
- Designed a secure, centralized cloud-based architecture, successfully replacing fragmented Excel-based workflows with automated, scalable data pipelines.
- Implemented ISO 27001-aligned policies and Role-Based Access Control (RBAC) to mitigate vendor data leakage risks.
- Optimized operational efficiency and PDPA compliance through centralized data authority, resulting in streamlined vessel tracking and enhanced business intelligence.

For a comprehensive overview of my work, please visit my : <https://jayashinijayaweera.netlify.app/>

Singer Sri Lanka PLC |

June 2025 – July 2025

- The project evaluated four core operational areas at Singer Sri Lanka PLC, including order handling, delivery coordination, inventory replenishment, and warranty claims, to identify bottlenecks and inefficiencies.
- The research leveraged Business Process Model and Notation (BPMN) to visualize as-is workflows and proposed comprehensive digital transformations, such as workflow automation and ERP integration, to reduce processing times and enhance service delivery
- The project recommended data-driven improvements expected to optimize operational efficiency, minimize manual errors, and elevate overall customer satisfaction standards.

Pub/Sub | *Python*

June 2026

- Designed and implemented a Publish/Subscribe (Pub/Sub) messaging system in Python, enabling asynchronous, topic-based communication between multiple publishers and subscribers through a broker-based middleware architecture.
- Architected a multi-server failover system with broker instances running across multiple ports, implementing automatic client reconnection logic to eliminate single points of failure and improve system availability.
- Built a distributed messaging framework demonstrating core concepts of fault tolerance and failover handling, while identifying and addressing trade-offs between system availability and message reliability in non-replicated broker environments.

QA & Testing - FiremeX | *Personal Project / Self-Initiated*

June 2026 – Present

- Designed a comprehensive test strategy for the FiremeX AI-powered fire detection system, establishing critical quality benchmarks for CCTV platform reliability.
- Orchestrated full-lifecycle QA processes including functional, integration, performance, and security testing to ensure compliance with a 95% true positive detection rate and 99.5% system uptime. Utilized industry-standard automation frameworks like Selenium, Playwright, and Apache JMeter to execute automated regression suites and load testing, ensuring rapid and accurate validation of AI model retraining.

CERTIFICATIONS

Understand Threats & Prevent Attacks: Alison - CPD

Python for Beginners: University of Moratuwa

Cyber Security & Data: IBM SkillsBuild

Earned a Credential - Getting Started Cyber Security: IBM SkillsBuild

Understanding Manual Testing: LinkedIn Learning Community

Learning Go: LinkedIn Learning Community

Burp Suite Essential Training: LinkedIn Learning Community

Designing RESTful APIs: LinkedIn Learning Community

LEADERSHIP & VOLUNTEERING

Secretary

Present

ACM Student Chapter of UCSC

Assistant Secretary

Rotaract Club of UCSC

2025 - 2026

Faculty Coordinator

LEO Club of UCSC

2025 - 2026